SDI: Structured Diversity Initialization for Foundation Model Pretraining

Anonymous submission

Abstract

Foundation models (FMs) have achieved remarkable success across many domains, but their pretraining remains computationally expensive and time-consuming. We investigate structured initialization as an approach to improving FM pretraining. To analyze its effect, we introduce local feature accessibility, which characterizes the number and strength of distinct input directions linearly coupled to local output features. Motivated by this perspective, we propose Structured Diversity Initialization (SDI), which introduces activation-dependent structures into selected projection modules to enrich their initial feature spaces. We provide theoretical and empirical analyses of how SDI improves local feature accessibility and conduct systematic ablations to identify effective insertion positions and layer coverage. We evaluate SDI across language, vision, and scientific FMs, including LLaMA and Qwen2.5 models ranging from 71M to 6B parameters, Vision Transformer (ViT), WaveFormer, and Transolver. SDI consistently improves task-specific evaluation metrics over the default truncated Gaussian initialization. Furthermore, the results suggest favorable scaling behavior across the evaluated LLMs, with more pronounced gains at larger model scales and a 14.21% reduction in final validation perplexity on Qwen2.5-6B. By improving local feature accessibility, SDI enables more input directions to contribute to the initial feature representation, thereby accelerating pretraining with only minor parameter overhead.

Introduction

Foundation models (FMs) are large-scale models trained on broad data and adaptable to diverse downstream tasks across multimodal domains (Bommasani et al. 2022). FMs have achieved remarkable success across domains such as protein structure prediction (Jumper et al. 2021), chatbots (Brown et al. 2020), and scientific agents (M. Bran et al. 2024). However, training these models remains extremely expensive, often requiring massive compute budgets and prolonged optimization (Kaplan et al. 2020; Rae et al. 2021; Hoffmann et al. 2022). This challenge has motivated extensive work on improving training efficiency through better initialization (Glorot and Bengio 2010), normalization (Sun et al. 2026), architecture design (Wang et al. 2020), and optimization methods (Liu et al. 2025). Among these directions, initialization is particularly attractive because it can improve optimization with only minor additional parameter overhead.

Structured first-layer initialization (SFLI), a recently proposed method, has shown concrete evidence in accelerating the training process in several neural architectures, including MLPs, CNNs, ResNets, and PDE-oriented models such as PirateNets (Yang, Zhao, and Zhu 2024; Tang et al. 2025). The main idea of SFLI stems from spectral methods: it treats the first-layer neuron functions of a deep neural network (DNN) as basis functions for function approximation. Mirroring the principle that spectral basis functions should be as linearly independent as possible to obtain full representational power, SFLI artificially enhances the linear independence of a DNN's first-layer neuron functions, thereby increasing the capacity of the initial representation space spanned by these neuron functions (Yang, Zhao, and Zhu 2024; Tang et al. 2025). Despite these promising results, its role in FM pretraining remains largely unexplored. In particular, it is unclear whether SFLI is effective for FM pretraining, where it should be applied within the Transformer (Vaswani et al. 2017) and MLP blocks, and how its effect depends on projection type, activation variant, and layer coverage.

In this work, we advance the underlying theory of SFLI and propose Structured Diversity Initialization (SDI), a strategy that improves and extends SFLI for accelerating FM pretraining. We first evaluate SDI on representative Large Language Models (LLMs) families, including LLaMA (Touvron et al. 2023) and Qwen (Bai et al. 2023). We then extend the evaluation to vision and scientific foundation models based on ViT (Dosovitskiy et al. 2021), WaveFormer (Shu et al. 2026), and Transolver (Wu et al. 2024a). Our extensive experiments show that SDI consistently accelerates the pretraining process and outperforms default FM initializations. These gains are observed across language, vision, and scientific foundation models with negligible compute overhead.

In addition, we conduct comprehensive ablation studies to identify the most effective way to apply SDI in FMs. Specifically, we study its insertion positions and layer coverage in FMs. We compare applying SDI to different subsets of projections in both attention and MLP blocks, and analyze which configurations are most effective. Besides, we also compare SDI with other strong initialization baselines in FMs, includ-

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ing Xavier (Glorot and Bengio 2010), Kaiming (He et al. 2015), and truncated Gaussian (TG) initialization (Devlin et al. 2019).

Our main contributions can be summarized as follows:

- We propose Structured Diversity Initialization (SDI), an initialization strategy that improves local feature accessibility in FM projection modules and accelerates pretraining.
- We provide theoretical analysis and empirical evidence showing that SDI strengthens the contribution of local input directions to output features at initialization.
- We demonstrate the effectiveness and scalability of SDI across language, vision, and scientific FMs, with consistent performance gains and minor parameter overhead.

Related Work

Pretraining Challenges of Foundation Models

FM pretraining requires massive datasets, accelerator clusters, and long training runs over trillions of tokens or samples, incurring substantial computational, financial, and energy costs (Sharir, Peleg, and Shoham 2020; Cottier et al. 2024). It can also suffer from unstable training dynamics, including loss spikes (Nishida, Nishida, and Saito 2024) and limited contributions from deep layers (Sun et al. 2025). Existing efforts address these challenges from different angles: compute-optimal scaling improves data–model allocation (Hoffmann et al. 2022), FlashAttention reduces attention I/O overhead (Dao et al. 2022), and LayerNorm Scaling stabilizes deep LLM training (Sun et al. 2025). Orthogonal to these approaches, initialization-based methods seek to improve optimization from the beginning of training.

Parameter Initialization

Parameter initialization significantly affects the learning outcome of FMs. Classical variance-preserving initialization schemes provide the foundation for stable DNN training. Xavier initialization (Glorot and Bengio 2010) balances the fan-in and fan-out of each layer to preserve forward activations and backward gradients, and is particularly suited to symmetric activations such as $\tanh$ and sigmoid. Kaiming initialization (He et al. 2015) adapts this principle to ReLU-type activations by compensating for the variance reduction caused by rectification, making it effective for very deep CNNs and residual networks. In large Transformer-based models, weights are also commonly initialized from Gaussian or TG distributions with carefully chosen variance, making the initialization scale a central hyperparameter for stable pretraining (Vaswani et al. 2017; Touvron et al. 2023; Bai et al. 2023).

Methodology

Preliminary: Pre-LN Transformer Block

We use the Pre-LN Transformer block in Figure 1 as the reference architecture, as it is a common core module in modern FMs. Given hidden states X_ℓ, the block can be summarized as

$$X'_\ell = X_\ell + \mathrm{Attn}(\mathrm{LN}(X_\ell)),$$
$$X_{\ell+1} = X'_\ell + \mathrm{FFN}(\mathrm{LN}(X'_\ell)).$$

The candidate SDI insertion positions P_1–P_7 correspond to the query, key, value, output, gate, up, and down projections, respectively. Detailed definitions of the attention and gated FFN modules are provided in the Technical Appendix.

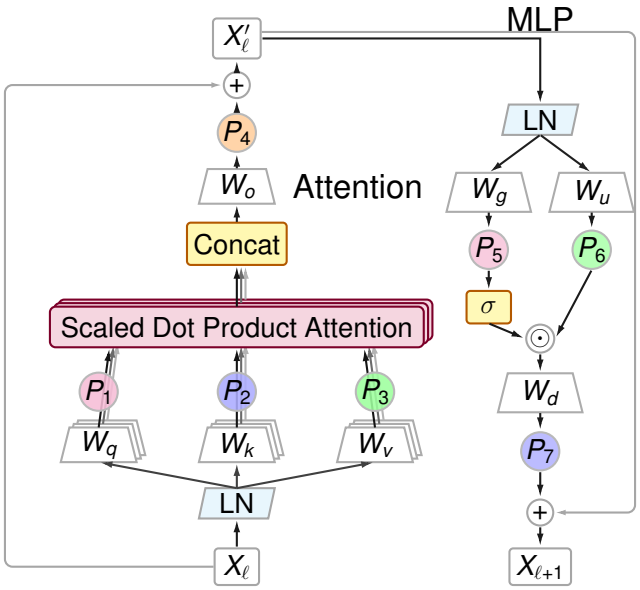


Figure 1: Pre-LN Transformer block with candidate SDI insertion positions P_1–P_7.

SDI initialization strategy

This section presents the motivation, mechanism, and implementation of the proposed SDI strategy in FMs. A simple example illustrates the underlying motivation. Let

$$z = (z_1, z_2, z_3)^\top \sim \mathcal{N}(0, I_3), \quad \phi(z) = (z_1, z_2, z_1 + z_2)^\top.$$

Although z is three-dimensional, $\phi(z)$ is independent of z_3. Consequently, no downstream predictor that depends only on $\phi(z)$ can represent a target that varies with z_3. For example,

$$\inf_g \mathbb{E}\Big[\big(z_3 - g(\phi(z))\big)^2\Big] = \mathbb{E}[\mathrm{Var}(z_3 \mid \phi(z))] = 1.$$

Thus, inaccessible input directions can create a local representational bottleneck. Prior work has similarly associated rank-deficient intermediate representations with information loss and impaired gradient propagation in deep networks and Transformers (Daneshmand et al. 2020; Noci et al. 2022).

Motivated by this observation, we propose SDI, an initialization strategy designed to increase the number and strength of local input directions that are linearly coupled to neuron outputs at initialization. Applying this idea to FMs is challenging because modern architectures typically contain stacks of Transformer blocks. We therefore decompose each block into its projection modules and identify P_1–P_7 in Figure 1 as candidate SDI insertion positions.

We next introduce a local measure of this input–output coupling. For a projection module p in layer ℓ, let $z \sim \mu_{\ell,p}$ denote its hidden-state input and let $\phi(z) \in \mathbb{R}^n$ denote its local feature output. Because we analyze one module at a time, we omit (ℓ, p) and write $z \sim \mu$, where $z \in \mathbb{R}^d$. We assume that μ is symmetric about the origin and has nonsingular covariance and finite fourth moment:

$$\mathbb{E}[z] = 0, \qquad \Sigma := \mathbb{E}[zz^\top] \succ 0, \qquad \mathbb{E}\|z\|_2^4 < \infty. \quad (1)$$

Definition 1 (Input–feature coupling matrix)**.** The input–feature coupling matrix of ϕ under μ is

$$T(\phi) = \frac{1}{\sqrt{n}} \mathbb{E}\big[\phi(z) z^\top\big]\, \Sigma^{-1/2}, \quad (2)$$

where $\Sigma^{-1/2}$ is the symmetric inverse square root of Σ.

Because $\mathbb{E}[z] = 0$, the expectation in Equation (2) is the cross-covariance between $\phi(z)$ and z. Let

$$\widetilde{z} = \Sigma^{-1/2} z, \qquad \widetilde{\phi}(z) = \phi(z) - \mathbb{E}[\phi(z)].$$

Then

$$T(\phi) = \frac{1}{\sqrt{n}} \mathbb{E}\Big[\widetilde{\phi}(z)\widetilde{z}^\top\Big],$$

so $T(\phi)$ removes the scale and correlation structure of the local input while retaining its coupling to the output features.

To see the role of its singular values, let

$$T(\phi) = U \operatorname{diag}(s_1, \ldots, s_r) V^\top$$

be its singular value decomposition, with singular-vector pairs (u_j, v_j). Then

$$\frac{1}{\sqrt{n}} \mathbb{E}\Big[\big(u_j^\top \widetilde{\phi}(z)\big)\big(v_j^\top \widetilde{z}\big)\Big] = s_j(T). \quad (3)$$

Hence, $s_j(T)$ measures the linear coupling strength between the whitened input direction v_j and the feature direction u_j. Different singular-vector pairs identify mutually orthogonal coupling modes. We therefore define the local ϵ rank by

$$r_\epsilon(T) = \#\{j : s_j(T) > \epsilon\}, \qquad \epsilon > 0, \quad (4)$$

which counts the input directions that contribute to the local output features with coupling strength above ϵ. This provides a metric for measuring the *Local Feature Accessibility*: the part of the feature map that is linearly accessible from the local input.

For the motivating example,

$$T(\phi) = \frac{1}{\sqrt{3}} \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 1 & 1 & 0 \end{pmatrix},$$

whose singular values are 1, $1/\sqrt{3}$, and 0. Therefore, $r_\epsilon(T) = 2$ for $0 < \epsilon < 1/\sqrt{3}$, correctly identifying that only the z_1 and z_2 input directions are accessible from the feature output. In the Analysis Section, we will demonstrate empirically and theoretically that SDI indeed improves the local feature accessibility.

Implementation

This section formally introduces SDI and how to apply it in FM architectures.

We begin by defining one of the most common structures in FMs, the *linear transform*. Suppose that $\Omega \subseteq \mathbb{R}^{d_{\text{in}}}$ is the input domain. A linear transform is a function $\Phi : \Omega \to \mathbb{R}^{d_{\text{out}}}$ such that $\Phi(\mathbf{x}) = \mathbf{W}\mathbf{x} + \mathbf{b}$, where $\mathbf{W} \in \mathbb{R}^{d_{\text{out}} \times d_{\text{in}}}$ and $\mathbf{b} \in \mathbb{R}^{d_{\text{out}}}$. Given a nonlinear activation function σ, SDI can be viewed as an operator $S_\sigma : \mathcal{L}(\Omega, \mathbb{R}^{d_{\text{out}}}) \to \mathcal{N}_\sigma(\Omega, \mathbb{R}^{d_{\text{out}}})$, which maps a linear transform to a structured nonlinear transform at initialization. Specifically,

$$\begin{aligned} S_\sigma(\Phi)(\mathbf{x}) &= S_\sigma(\mathbf{W}\mathbf{x} + \mathbf{b}) \\ &= \sigma(\gamma(\mathbf{A}\mathbf{x} + \mathbf{c}))\,, \end{aligned} \quad (5)$$

where $\gamma > 0$ is a scaling coefficient whose selection depends on experience. Here, $\mathbf{A} = [\boldsymbol{\alpha}_1, \ldots, \boldsymbol{\alpha}_{d_{\text{out}}}]^\top \in \mathbb{R}^{d_{\text{out}} \times d_{\text{in}}}$, with $\boldsymbol{\alpha}_i \in \mathbb{R}^{1 \times d_{\text{in}}}$, is a specially initialized trainable matrix whose construction depends on the choice of nonlinear activation σ. $\mathbf{c} \in \mathbb{R}^{d_{\text{out}}}$ is a bias vector initialized such that the components c_i are set to be well-separated on Ω to keep the neuron functions linearly independent. The i-th component of the initialized transform is therefore

$$[S_\sigma(\Phi)(\mathbf{x})]_i = \sigma(\gamma(\boldsymbol{\alpha}_i \mathbf{x} + c_i))\,.$$

Here, $\boldsymbol{\alpha}_i$ denotes the i-th row of $\mathbf{A}$, and c_i denotes the corresponding bias component. The above describes the general formulation of SDI, while its exact application in FMs should be adapted to the specific module structure. We therefore describe its implementation with reference to Figure 1.

SDI Applied to the Attention Block When applied to positions P_1–P_4, SDI initializes the query, key, value, and output projections W_Q, W_K, W_V, W_O, where $Q = W_Q X$, $K = W_K X$, and $V = W_V X$. We empirically find that cosine is the most effective activation for these projections and therefore set $\sigma(x) = \cos(x)$ in Eq. (5). We sample $A_{ij} \sim \mathcal{N}(0, \tau^2)$, select τ empirically, and set

$$\gamma = C \frac{n^{1/d} - 1}{|\Omega|^{1/d}}, \qquad C \in [0.5, 2],$$

where n is the layer width, d is the input dimension, Ω is the input domain, and C is selected empirically. The formula of γ is an empirical heuristic developed for low-dimensional DNNs; for FMs, we calibrate γ around the value it suggests. The bias components are initialized as the arithmetic sequence $c_i = c_0 + (i-1)\Delta_c$, where Δ_c is chosen on the same scale as τ. Therefore, the i-th component of an SDI-initialized attention projection is

$$[S_{\cos}(\Phi)(\mathbf{x})]_i = \cos(\gamma(\boldsymbol{\alpha}_i \mathbf{x} + c_i))\,, \qquad \alpha_{ij} \sim \mathcal{N}(0, \tau^2).$$

SDI Applied to the MLP Block At positions P_5–P_7, SDI is applied to the projections within the SwiGLU block,

$$H(X) = \text{SiLU}(W_g X) \odot W_u X, \quad \text{MLP}(X) = W_d^\top H(X).$$

Applying SDI to the gate, up, and down projections, respectively, yields

$$\begin{aligned} P_5: &\quad W_d^\top\, [\text{SiLU}(\gamma(AX + c)) \odot W_u X]\,, \\ P_6: &\quad W_d^\top\, [\text{SiLU}(W_g X) \odot \gamma(AX + c)]\,, \\ P_7: &\quad \gamma(AH(X) + c)\,. \end{aligned}$$

Thus, P_5 retains the native SiLU activation of the gate branch, whereas P_6 and P_7 use identity activation. For each position, A, γ, and c follow the initialization rules described for the attention block, with dimensions adapted to the corresponding projection.

Experiments

Experimental setups

We evaluate SDI across three FM domains—language, vision, and scientific computing—using C4 (Raffel et al. 2019), ImageNet-1K (Liu et al. 2023), and representative industrial PDE datasets (Li et al. 2020, 2022). These widely adopted benchmarks are chosen for their task difficulty and comparability with prior work. To isolate the effect of initialization, we follow the training configurations of prior strong baselines in each domain and change only the initialization strategy. Detailed hyperparameters and training settings are provided in the *Detailed Configurations of Main Experiments* section of the Technical Appendix.

Evaluation metrics follow the standard conventions of each domain. For LLMs, we report pretraining perplexity (PPL), measured on a held-out validation set during pretraining. PPL is computed as $\exp(\mathcal{L})$, where $\mathcal{L}$ is the validation cross-entropy loss, and measures the model's uncertainty in next-token prediction. For vision foundation models, we report image classification accuracy on the validation set, the standard metric for ImageNet-1K classification. For scientific foundation models, we report the mean and standard deviation of relative L^2 error on the test set, the standard metric for PDE learning tasks.

LLM Pretraining: LLaMA and Qwen

To evaluate the efficacy of SDI in LLM pretraining, we benchmark it against the default TG initialization used in LLaMA and Qwen architectures. Following the experimental setup of Li, Yin, and Liu (2024), we establish baselines for LLaMA models ranging from 71M to 1B parameters using TG initialization with either Pre-LN (Dai et al. 2019) or Mix-LN (Li, Yin, and Liu 2024) normalization methods. Subsequently, we apply SDI at position P_5 across all layers, using Pre-LN and Mix-LN as the underlying normalization methods. The rationale for applying SDI at position P_5 is based on the top-3 best-performing SDI insertion positions summarized in Table 5, with further experiments to select the optimal application position among these top-3 positions.

All models are trained from scratch on the English portion of the C4 dataset (Raffel et al. 2019), optimized on the training split and evaluated on the validation split, following Li, Yin, and Liu (2024). We employ the Adam optimizer with a cosine learning-rate schedule and a minimum learning-rate ratio of 0.1. To ensure a fair comparison, all training hyperparameters remain identical across models of the same size, isolating the initialization method as the sole variable.

The results are shown in Table 1. SDI consistently outperforms the baselines across LLaMA models of varying scales. For each model size, we compare the best-performing SDI configuration against the strongest TG baseline. On LLaMA-1B, SDI reduces PPL by 7.73%, demonstrating its effectiveness in accelerating LLaMA pretraining. On the remaining LLaMA models, SDI reduces PPL by more than 0.3 points relative to the strongest baseline. The relative improvement is substantially larger on LLaMA-1B than on the smaller models (71M–250M), suggesting favorable scaling behavior. Moreover, SDI consistently outperforms the corresponding TG baseline under both Pre-LN and Mix-LN, indicating its robustness and compatibility with different normalization schemes.

Overall, applying SDI to the appropriate insertion positions in LLaMA models can notably reduce PPL and accelerate pretraining. However, the insertion positions need to be selected carefully to obtain optimal acceleration benefits. A comprehensive analysis of the optimal insertion positions for SDI is provided in the ablation study.

Evaluation with Qwen We further assess the compatibility of SDI with other LLM architectures by applying it to Qwen2.5, a series of strong LLMs developed by Alibaba (Qwen et al. 2024). We train the Qwen2.5 models on C4 using training settings consistent with those of the LLaMA experiments, with model-size-specific hyperparameters provided in the Technical Appendix. We compare SDI with the default TG initialization. SDI is applied to position P_5 in Figure 1, one of the top-3 insertion positions in Table 5. The results in Table 2 show that SDI consistently reduces PPL across all evaluated Qwen2.5 model sizes (200M–0.5B), with absolute reductions of 0.56–1.01 PPL points. The consistent performance gains observed across model scales and architectures show that SDI holds promising potential for pretraining acceleration and PPL reduction in LLMs.

Scaling Up to 6B Model Evaluating whether the acceleration benefits of SDI scale to larger models, such as a 6B-parameter LLM, is critical for determining its viability in industrial-scale LLM pretraining. To this end, we evaluate SDI on the Qwen2.5-6B architecture. We train Qwen2.5-6B on C4 using a training configuration nearly identical to that used in the preceding Qwen2.5-0.5B experiments. Due to computational constraints, we train the model for $13{,}000$ steps and report the evaluation PPL curves in Figure 2. The results show that SDI consistently outperforms TG initialization throughout training, with a substantial 44.39% PPL reduction at the early stage and a 14.21% reduction at the final step.

Across the above experiments, SDI shows a clear scaling behavior: its relative reduction in final evaluation PPL generally increases with model size (Figure 2). In particular, SDI achieves a 14.21% reduction on Qwen2.5-6B and a 7.73% reduction on LLaMA-1B over the TG baseline, indicating that SDI becomes increasingly effective at larger scales.

Vision Foundation Model Pretraining: ViT and WaveFormer

To evaluate the generalizability of SDI to vision FMs, we conduct experiments on ViT and WaveFormer. For ViT, we apply SDI to position P_3 in Figure 1 across all layers, as it is one of the top-3 positions in Table 5. WaveFormer replaces self-attention with a wave propagation operator for modeling long-range token dependencies, and therefore is not directly

Model	LLaMA-71M	LLaMA-130M	LLaMA-250M	LLaMA-1B
Pre-LN + TG	34.63	27.13	22.04	17.33
Mix-LN + TG	34.30	26.60	21.94	17.58
Pre-LN + SDI	34.03	26.46	**21.30**	**15.99**
Mix-LN + SDI	**33.94**	**26.05**	21.55	16.08

Table 1: Final evaluation PPL (↓) of LLaMA models under Pre-LN and Mix-LN normalization methods. **Bold** indicates the best results. TG denotes truncated Gaussian initialization.

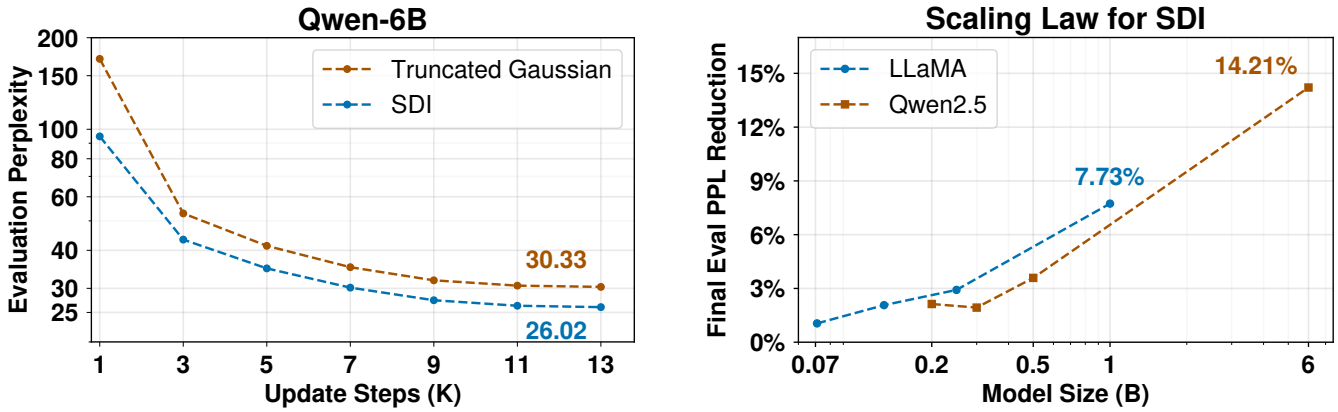


Figure 2: Pretraining performance and scaling behavior of SDI. Left: Evaluation PPL curves of Qwen2.5-6B under SDI and truncated Gaussian initialization, with a log-scaled y-axis. Right: Scaling behavior of SDI across LLaMA and Qwen2.5 model sizes.

Model	Qwen-200M	Qwen-300M	Qwen-0.5B
TG	35.19	28.99	28.15
SDI	**34.44**	**28.43**	**27.14**

Table 2: Comparison of final evaluation PPL (↓) between SDI and truncated Gaussian (TG) initialization on Qwen2.5 models.

described by Figure 1. We use WaveFormer to test whether SDI generalizes beyond self-attention-based FMs. For WaveFormer, SDI is applied to all linear layers. All models are pretrained on ImageNet-1K (Liu et al. 2023) for 120 epochs and compared with the default TG initialization, while all other settings follow Liu et al. (2022). The results in Table 3 show that SDI consistently improves image classification accuracy across all models, with a notable accuracy gain of 1.3 on ViT-Small. SDI also yields small but consistent gains on both WaveFormer variants. These results demonstrate that SDI generalizes effectively to vision FMs, including architectures beyond self-attention.

Scientific Foundation Model Pretraining: Transolver

Transolver is a transformer-based neural solver for PDEs on complex geometries, which learns intrinsic physical states instead of relying on traditional grid-based methods (Wu et al. 2024b). It excels at large-scale industrial simulations such as car and airfoil design (Wu et al. 2024b). To validate

Model	ViT-T	ViT-S	WaveFormer-T	WaveFormer-S
TG	68.65	76.50	79.98	82.13
SDI	**69.02**	**77.80**	**80.05**	**82.16**

Table 3: Comparison of image classification accuracy (↑) between SDI and truncated Gaussian (TG) initialization on ViT and WaveFormer. T and S denotes Tiny and Small, respectively.

SDI on Transolver, we evaluate it on three standard 2D structured mesh benchmarks: Airfoil, Pipe, and Plasticity (Li et al. 2020, 2022). Strictly adhering to the training configurations in Wu et al. (2024b), we apply SDI to all linear layers within the Transolver architecture. As shown in Table 4, SDI generalizes effectively to Transolver, delivering performance on par with or slightly better than the default TG initialization. Results report the mean and standard deviation of the relative L^2 test error over three independent runs.

Ablation Study

Each ablation uses the corresponding main-experiment setup for the same model (e.g., LLaMA-130M or Qwen2.5-200M), with only minor changes for computational convenience.

Ablation on SDI Insertion Positions We first study how the insertion position of SDI affects FM pretraining, using LLaMA-130M as an example. As shown in Table 5, we consider two module families: attention blocks, where SDI is

Model	AIRFOIL	PIPE	PLASTICITY
TG	$5.72e$-$3 \pm 2.53e$-4	$4.57e$-$3 \pm 1.40e$-4	$1.31e$-$3 \pm 3.27e$-5
SDI	$5.71e$-$3 \pm 9.49e$-5	$4.43e$-$3 \pm 7.67e$-5	$1.30e$-$3 \pm 2.16e$-5

Table 4: Comparison of relative L^2 test error (↓) between SDI and truncated Gaussian (TG) initialization on the Transolver model.

Attention Blocks			MLP Blocks		
Position(s)	Activation Function	Eval PPL (↓)	Position(s)	Activation Function	Eval PPL (↓)
P_1	cos	25.602	P_6	SiLU	25.182
P_2	cos	25.215	P_5	SiLU	**25.018**
P_3	cos	**24.947**	P_7	SiLU	25.181
$P_1 + P_2$	cos	26.568	$P_5 + P_6$	SiLU	25.279
$P_1 + P_3$	cos	25.014	$P_6 + P_7$	SiLU	25.876
$P_2 + P_3$	cos	25.142	$P_5 + P_7$	SiLU	25.042
$P_1 + P_2 + P_3$	cos	26.972	$P_5 + P_6 + P_7$	SiLU	25.128

Table 5: Ablation results for different SDI insertion positions in all decoder layers of LLaMA-130M. The position notation P_1–P_7 follows Figure 1. The best result within each block type is highlighted in bold.

applied to different subsets of the query (P_1), key (P_2), and value (P_3) projections; and MLP blocks, where SDI is applied to different subsets of the gate (P_5), up (P_6), and down (P_7) projections. For each family, we evaluate all single, pairwise, and full combinations. Unless otherwise specified, these configurations are applied to the corresponding projections in all decoder layers. In the *Supplementary Experiments* section of Technical Appendix, we compare several activations for attention-side SDI and find that cosine activation performs best; therefore, we fix cosine activation and compare only insertion positions. Table 5 shows that the best attention-side and MLP-side insertion positions are P_3 and P_5, corresponding to the value projection and gate projection, respectively.

We further evaluate the combination $P_3 + P_5$ to test whether the best attention-side and MLP-side insertions are complementary. This combination achieves the best overall performance with an evaluation PPL at 24.605, indicating that the two insertions are complementary rather than redundant. We therefore use $P_3 + P_5$ as the default SDI insertion pattern in the subsequent layer-coverage ablations. In the main experiments, we adopt the top-3 configurations from this ablation: P_3, P_5, and $P_3 + P_5$.

Effect of Varying the Number of Decoder Layers Initialized with SDI. We next study the effect of varying the number of decoder layers initialized with SDI in LLaMA-130M. Based on the last section, $P_3 + P_5$ yields the best performance. We apply SDI to this optimal combination in each decoder layer and vary the layer coverage. Specifically, we consider six settings: applying SDI to only the first decoder layer, and to layers 0–2 (25%), layers 0–3 (33%), layers 0–5 (50%), layers 0–7 (66%), and layers 0–11 (100%). We compare these settings using final evaluation PPL. As shown in Table 6, increasing layer coverage generally improves SDI performance, with PPL decreasing from 26.604 when only layer 0 is initialized to 24.605 when all layers are initialized.

L0 only	L0–2 (25%)	L0–3 (33%)	L0–5 (50%)	L0–7 (66%)	L0–11 (100%)
26.604	25.884	25.415	24.937	24.660	**24.605**

Table 6: Comparison of the final evaluation PPL (↓) of LLaMA-130M with varying numbers of decoder layers initialized with SDI. L denotes decoder layers.

Model	Kaiming	Xavier	TG	SDI
Qwen-200M	39.85	35.32	35.19	**34.44**

Table 7: Comparison against other initialization methods on Qwen2.5-200M. TG denotes truncated Gaussian initialization.

Comparing with Other Important FM Initialization Methods Using Qwen2.5-200M, we further compare SDI with widely used FM initialization methods, including Xavier and Kaiming. Results from Table 7 indicate that TG and Xavier achieve similar performance, while both substantially outperform Kaiming. However, all of them fail to reduce PPL below 35, falling short of SDI. This result demonstrates the advantage of SDI over other important initialization methods in FMs.

Analysis

This section empirically shows that SDI improves local feature accessibility, i.e., the number and strength of local input directions linearly coupled to neuron outputs at initialization, and provides a theoretical explanation for this effect.

We analyze the LLaMA-71M setting in Table 1, where

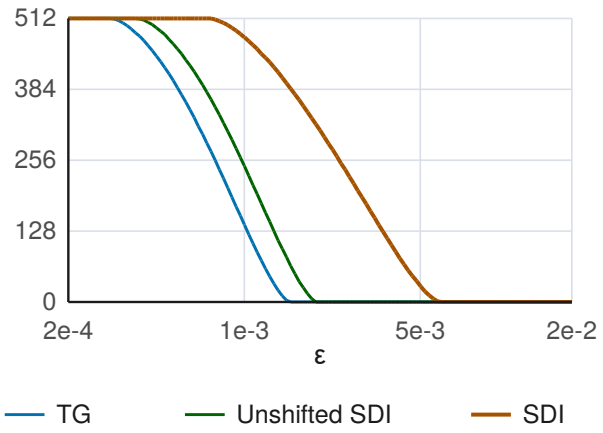


Figure 3: Local ϵ-rank of the LLaMA-71M SwiGLU feature map under different initializations. Higher values indicate more accessible input directions. TG denotes truncated Gaussian.

SDI is applied to the gate projection P_5. Given a local input $z \in \mathbb{R}^d$, the corresponding feature map is

$$\phi(z) := \mathrm{SiLU}(W_g z + b_g) \odot W_u z \in \mathbb{R}^n, \tag{6}$$

where $W_g, W_u \in \mathbb{R}^{n\times d}$, $b_g \in \mathbb{R}^n$, and $\odot$ denotes element-wise multiplication. We compare three gate initializations:

$$\begin{aligned} \text{TG:} \quad & W_g \sim \mathrm{TruncNormal}(0, \sigma_g^2), \quad b_g = 0, \\ \text{unshifted SDI:} \quad & W_g = \gamma A, \quad b_g = 0, \\ \text{SDI:} \quad & W_g = \gamma A, \quad b_g = \gamma c. \end{aligned}$$

Using the same nominal gate scale, $\sigma_g = \gamma = 0.02$, we compute the local ϵ-rank of Eq. (6) under each initialization. All methods share the same W_u and inputs. Further details are provided in the Supplementary Materials. As shown in Figure 3, all methods attain full rank for sufficiently small ϵ, but SDI retains substantially more accessible directions as the threshold increases. At $\epsilon = 10^{-3}$, the median ranks are 140, 246, and 478 for TG, unshifted SDI, and SDI, respectively. These results show that the structured shift in SDI substantially strengthens local input–feature coupling at initialization.

Why SDI Improves Local Feature Accessibility

Let

$$\begin{aligned} \phi_0(z) &= \mathrm{SiLU}(\gamma A z) \odot W_u z, \\ \phi_c(z) &= \mathrm{SiLU}(\gamma(Az + c)) \odot W_u z. \end{aligned} \tag{7}$$

denote the feature maps induced by unshifted and shifted SDI, respectively. We write $T_0 = T(\phi_0)$ and $T = T(\phi_c)$ for their corresponding input–feature coupling matrices. Under the symmetry and moment assumptions preceding Eq. 1, we obtain the following result.

Theorem 1 (First-order effect of the SDI shift)**.** *As $\gamma \to 0$, with μ, A, W_u, and c fixed,*

$$\begin{aligned} T_0 &= O(\gamma^2), \\ T &= \frac{\gamma}{2\sqrt{n}} \mathrm{diag}(c) W_u \Sigma^{1/2} + O(\gamma^2). \end{aligned} \tag{8}$$

The remainder terms are measured in spectral norm, with constants independent of γ.

Theorem 1 establishes an order separation between unshifted SDI and SDI. Without the shift, the input–feature coupling appears only at order $O(\gamma^2)$. The structured shift introduces the first-order term $\frac{\gamma}{2\sqrt{n}} \mathrm{diag}(c) W_u \Sigma^{1/2}$, raising the coupling strength to $O(\gamma)$.

This order separation directly yields the following corollary.

Corollary 2 (Rank separation)**.** *Suppose $n \geq d$ and $B := \mathrm{diag}(c) W_u \Sigma^{1/2}$ has full column rank, with smallest singular value $s_d(B) > 0$. Then there exist constants $K > 0$ and $\gamma_0 > 0$ such that, for every $0 < \gamma < \gamma_0$,*

$$\begin{aligned} K\gamma^2 < \epsilon < \frac{\gamma}{2\sqrt{n}} s_d(B) - K\gamma^2, \\ r_\epsilon(T_0) = 0, \qquad r_\epsilon(T) = d. \end{aligned}$$

Corollary 2 provides a concrete mechanism by which SDI can improve local feature accessibility. Under the stated assumptions, the structured shift introduces an $O(\gamma)$ term that is absent from the zero-shift initialization. When $B = \mathrm{diag}(c) W_u \Sigma^{1/2}$ has full column rank, this term covers all d input directions, creating a range of thresholds ϵ for which $r_\epsilon(T) = d$ while $r_\epsilon(T_0) = 0$. TG initialization is also small-scale and zero-bias; under the same symmetry assumption, its $O(\gamma)$ term vanishes, so its input–feature coupling begins at $O(\gamma^2)$, as for T_0. This suggests why SDI can yield greater local feature accessibility than TG initialization. Complete proofs are provided in the Supplementary Materials.

Conclusion

We propose Structured Diversity Initialization (SDI), an initialization strategy that improves local feature accessibility in selected FM projection modules. Through activation-dependent structured shifts, SDI increases the number and strength of local input directions linearly coupled to output features at initialization. Across language, vision, and scientific FMs of different scales, SDI consistently improves task-specific evaluation metrics over default initialization, while introducing only minor parameter overhead. These results demonstrate the effectiveness and generalizability of SDI for accelerating FM pretraining.

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